IEEE Access
Multidisciplinary ⁝ Rapid Review ⁝ Open Access Journal

Received 5 June 2025, accepted 1 July 2025, date of publication 3 July 2025, date of current version 10 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3585773

SURVEY

Applications of Machine Learning in Human Factors and Ergonomics: A Comprehensive Review of Research From the Past Decade

ERMAN ÇAKIT [1] AND WALDEMAR KARWOWSKI [2], (Life Senior Member, IEEE)
[1]Department of Industrial Engineering, Gazi University, 06570 Ankara, Türkiye
[2]Department of Industrial Engineering and Management Systems, University of Central Florida, Orlando, FL 32816, USA

Corresponding author: Erman Çakit (ecakit@gazi.edu.tr)

ABSTRACT This study provides a comprehensive literature review of 130 peer-reviewed journal articles published between 2015 and 2024 on the application of Machine Learning (ML) algorithms in Human Factors and Ergonomics (HFE). The selected studies cover eight key application areas, including Work-Related Musculoskeletal Disorders (WRMSDs), Mental Workload Assessment (MWLA), physical workload, and occupational safety. ML techniques examined include Support Vector Machines (SVMs), Artificial Neural Networks (ANNs), Random Forest (RF), Decision Trees (DTs), K-Nearest Neighbors (KNN), Naive Bayes (NB), boosting algorithms, and k-means clustering. Among the reviewed articles, Artificial Neural Networks (ANNs) and Deep Learning (DL) methods were the most prevalent, at 36%, followed by Support Vector Machines (SVMs) at 22% and Random Forests (RF) at 19%. Performance benchmarks from these studies indicate classification accuracies ranging from 83% to 97.4% for tasks such as posture recognition, MWLA, and fatigue detection. For instance, ANN models for Electroencephalography (EEG)-based MWLA achieved up to 94% accuracy, while SVM classifiers reached up to 97.94% in posture classification. KNN, when optimized for k-values, yielded accuracy rates exceeding 90% in ergonomic applications. There was a marked increase in ML adoption in HFE post-2019, with 2020 publications nearly doubling compared to 2019. This review not only benchmarks algorithmic performance but also identifies research gaps and provides guidance for future machine-learning applications in human factors, ergonomics, and human-centered systems design.

INDEX TERMS Ergonomics, human factors, machine learning, review.

I. INTRODUCTION

Human factors and ergonomics (HFE) is a unique discipline that focuses on the nature of human-artifact relationships. HFE aims to optimize the interactions between people, products, tools, and systems to enhance safety, well-being, and performance [1], [2]. The HFE field uses various research methods, data analysis, and human-centered design principles to identify and address human performance, workload, usability, and safety issues in various domains. The above often requires gathering, analyzing, and interpreting large amounts of experimental data to gain insightful information and provide useful system design knowledge. This can be achieved by the application of sophisticated algorithms borrowed from computer science, such as machine learning (ML), deep learning (DL), and other forms of artificial intelligence (AI).

ML refers to enabling "computers the ability to learn without being explicitly programmed" [3], i.e., providing a computer's ability to improve its performance in a specific task T, as measured by a performance metric P, by learning from experience E [4]. This definition captures the essence of ML, wherein algorithms are designed to learn from data and improve performance over time through experience. Regression, classification, clustering, and reinforcement learning tasks are all frequently accomplished by the application of ML algorithms.

The integration of ML in HFE presents significant opportunities for enhancing workplace safety, optimizing human-computer interaction, and improving overall

The associate editor coordinating the review of this manuscript and approving it for publication was Chao Tong.

efficiency. ML techniques enable ergonomic risk assessment by analyzing posture and motion data through computer vision and wearable sensors, helping prevent musculoskeletal disorders. In human-computer interaction, ML personalizes user experiences by adapting interfaces based on behavioral patterns, improving accessibility and usability. Fatigue and stress monitoring is another key area where ML-powered wearable devices analyze physiological data to detect early signs of cognitive overload, reducing risks in high-stakes environments such as healthcare and manufacturing. Additionally, predictive analytics helps foresee workplace accidents by analyzing historical safety data, allowing organizations to implement proactive safety measures. The integration of ML with robotics and automation enhances human-robot collaboration, particularly in industrial settings where AI-powered exoskeletons assist workers in physically demanding tasks. Recent trends include computer vision for ergonomic assessment [5], AI-driven wearables for real-time health monitoring [6], and natural language processing for analyzing workplace safety reports [7]. The adoption of ML in HFE offers numerous benefits, including improved workplace safety, increased productivity, cost reduction by preventing injuries, and personalized user experiences that enhance efficiency. By leveraging these advancements, industries can create safer, more adaptive, and efficient work environments.

Numerous ML algorithms, alone and in combination, have been used in various HFE-related applications and problem-solving domains. This paper discusses the application of ML algorithms in HFE over the last ten years (2015–2024). The study begins by presenting its context, outlining the objectives, and emphasizing the significance of ML algorithms in HFE, along with the necessity of a literature review. The research methods and details of the literature search according to the steps are outlined in Section II. Section three introduces ML concepts and various ML techniques. Key findings regarding ML algorithms in HFE are presented in Section IV. Section V presents applications of ML algorithms in the field of HFE. Section VI discusses the review's findings. Finally, Section VII summarizes the main conclusions and discusses study limitations and pertinent future research implications.

II. METHODS

A. RESEARCH QUESTIONS

The presented literature review was aimed at addressing the following research questions:

- What ML applications have been used in HFE, and how can they be grouped based on how they work, where they are used, and what they aim to do?
- What are the existing research gaps in using ML in HFE, and what future studies could help improve how well ML works in this field? Also, what challenges must be solved to use ML more easily and effectively in HFE research and practice?

The steps based on the selected research questions and the search strategy described below is shown in Figure 1.

B. SOURCES AND SEARCH METHODS

The literature search was conducted by using several repositories, specialized search terms, and Boolean operators, such as (''ergonomics'' OR ''human factors'' OR ''musculoskeletal disorders'' OR ''low back pain'' OR ''mental workload'' OR ''physical workload'') AND (''machine learning'' OR ''deep learning'' OR ''support vector machines'' OR ''random forest'' OR ''artificial neural networks'' OR ''decision trees'' OR ''k-means clustering algorithms'' OR ''k-nearest neighbors'' OR ''naive bayes'' OR ''boosting algorithms''). The databases used in this study were Web of Science, Science Direct, Google Scholar, IEEE Xplore, Pubmed, EBSCO HOST and Scopus. The procedure was carried out following the sequence of steps depicted in Figure 1.

C. INCLUSION AND EXCLUSION CRITERIA

The following criteria were used to filter identified sources i) articles in English, ii) articles published in peer-reviewed journals, iii) articles relevant to HFE combined with ML, iv) articles with full text available, and v) articles published on or before December 20, 2024 (the last search date). Unrelated books, book chapters, review/survey articles, and conference proceedings were excluded. Choosing high-quality, high-impact journals from reputable digital libraries is an effective approach to ensure that the selected studies have undergone thorough peer review and possess strong scientific validity. Additionally, assessing the originality of the proposed techniques, along with technical elements such as datasets and evaluation methods, helps validate the relevance and contribution of these studies to the advancement of knowledge in the field of HFE. Focusing solely on journal articles allowed for access to comprehensive research that offers an in-depth exploration of this topic.

D. ARTICLE SELECTION

A total of 993 articles have been initially identified from the selected databases. Duplicate articles from 367 journals were removed. English-language publications were then assessed for compliance with the exclusion criteria, primarily on the basis of reviewing the abstracts and, in some cases, the introductions. This step involved preliminary screening to identify articles not aligned with the specific focus or criteria of the study. After the initial assessment, 464 items were removed in the second round of screening. Thirty-two articles were rejected in the third round because of the availability of more recent versions of the same dataset; thus duplicate or redundant content was carefully considered. After the three phases of application of the exclusion criteria, 130 articles were ultimately included in the evaluation (Figure 2). This final selection emphasized original research reported through peer-reviewed scientific publications, given the scholarly nature of the sources included in our study.

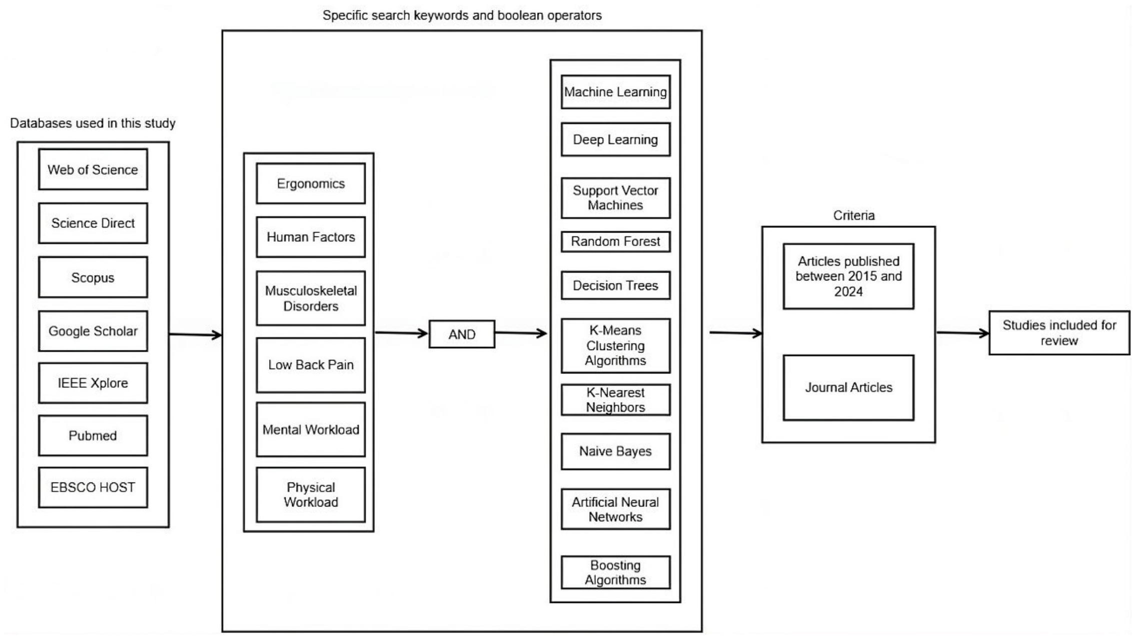

FIGURE 1. Databases and search keywords utilized in this literature review.

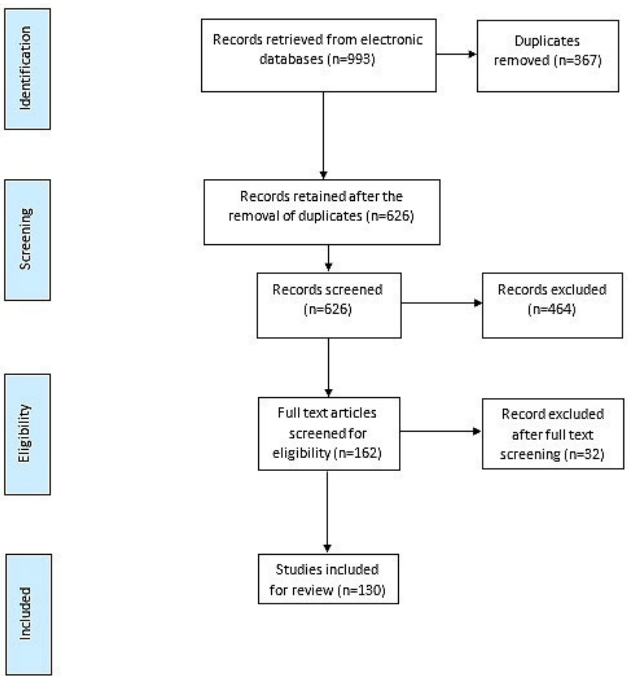

FIGURE 2. Steps for article selection process.

E. DIFFERENCES FROM PREVIOUS LITERATURE SURVEYS

While several literature surveys have been conducted in the recent past to analyze the integration of ML in the HFE field [8], [9], [10], [11], [12]. However, to the best of our knowledge, there has not been any comprehensive literature review on the application of ML methods in the field of HFE, which include boosting algorithms, and k-means clustering algorithms, k-nearest neighbors (KNN), naive Bayes (NB), decision trees, support vector machines (SVMs), random forest (RF), ANNs, and DL algorithms, have been published to date. Table 1 presents a comparison of the objectives and differences between previous literature surveys and the current study.

III. OVERVIEW OF ML

A. ML

ML is a learning process wherein algorithms are trained to learn patterns or relationships in a set of examples or training data. Each example is represented by a set of attributes or features [13]. These features can take different forms, such as numerical, categorical, binary, or ordinal [14]. During the training process, the algorithm learns to make predictions or classifications according to these features. The performance of the model is evaluated with a performance metric such as accuracy, precision, recall, or the F1 score. Various statistical and mathematical models, such as regression, DTs, ANNs, and SVMs, are commonly used in ML. After a model has been trained, it can be used to predict or classify new examples or data that it has not previously encountered. This process is called inference, and the model's performance on the testing data is used to evaluate its generalization performance. A general framework for ML algorithms is presented in Figure 3.

B. TYPES OF ML TECHNIQUES

There are four primary types of ML algorithms: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning [15], as shown in Figure 4.

TABLE 1. Past surveys on the applications of ML in HFE and their differences with the current review paper.

Review paper	Articles searched	Years	Aim	Comparison with the present review
Zhang et al., 2023 [8]	1141	2014-2021	Examine the current trends in research regarding the utilization of ML techniques within the realm of ergonomics.	✓ The benefits of ML applications for HFE not discussed. ✓ Papers published between 2021 and 2024 not included.
Chan et al., 2022 [9]	130	1990-2021	Explore the roles of ML methodologies in the proactive prevention of WMSDs.	✓ The review limited to WMSD prevention. ✓ Other application areas not covered. ✓ DL models not discussed. ✓ Papers published between 2021 and 2024 not included.
Stefana et al., 2021 [10]	28	2014-2020	Explore wearable devices recommended in academic research for ergonomic purposes and assess their potential to improve ergonomic applications.	✓ Focused on wearable devices only. ✓ DL models not discussed. ✓ The period 2020-2024 not included.
Gualtieri et al., 2021 [11]	55	2015-2018	Evaluate the status of collaborative robotics workcell design.	✓ Except human-machine interaction, other application areas not covered. ✓ DL models not discussed. ✓ Limited to papers published by 2018.
Lim and D'Souza, 2020 [12]	78	1998-2019	Examine the current and developing applications of wearable inertial sensing technology within the realm of occupational ergonomics, focusing on the assessment of biomechanical exposure during physical work tasks	✓ The main emphasis on the application of sensing technologies for monitoring biomechanical exposure in physical work settings. ✓ DL models not covered. ✓ Limited to papers published by 2019.

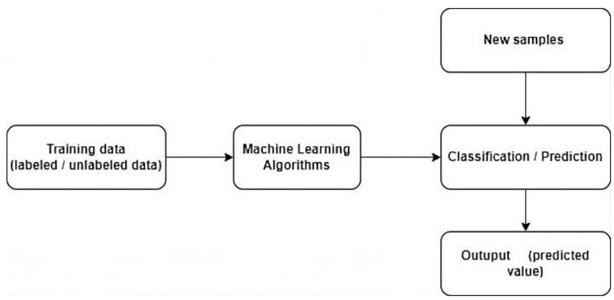

FIGURE 3. General framework for ML algorithms.

1) SUPERVISED LEARNING

Supervised learning algorithms are widely applicable and commonly used in various real-world problems. The goal of supervised learning is to learn a good approximation of the true mapping from the input vector to the output vector by using a dataset of labeled examples [17]. The most common supervised learning algorithms, such as ANNs, SVM, and DTs, aim to minimize a predefined cost or loss function to accurately predict new, previously unseen data. To do so, the labeled dataset is split into a "training" set and a "test" set: the former is used to train the model, and the latter is used to assess the prediction accuracy of the model. The output variables can be either categorical or continuous, thus resulting in classification or regression tasks, respectively [18]. In the case of categorical variables, classification tasks are used to predict the category or class of a given input, such as predicting the quality of a production batch on the basis of physical properties. For continuous variables, regression tasks are used to predict the numerical value of a given input, such as predicting the physical properties of an item according to its picture. The success of supervised learning relies on having a high quality labeled dataset that accurately represents the true mapping between the input vector and the output vector. Additionally, the effectiveness of different supervised learning algorithms depends on the specific problem being addressed and the complexity of the mapping between the input and output variables.

2) UNSUPERVISED LEARNING

Unsupervised learning is used with unlabeled datasets in which the output vector is missing. Instead of making predictions, the main goal of unsupervised learning is to detect and extract patterns in the data. This process can involve finding clusters of similar data or reducing the dimensionality of the data to identify key factors that affect the data points [19]. Unsupervised learning is often used in exploratory data analysis and can be used for tasks such as clustering, feature extraction, and dimensionality reduction [16]. This method is sometimes referred to as descriptive learning, because it focuses on describing the data rather than making predictions or classifications.

3) SEMI-SUPERVISED LEARNING

Whereas supervised learning uses labeled data to learn from examples and map input to output, unsupervised learning does not use labels, and is used to find patterns and structures

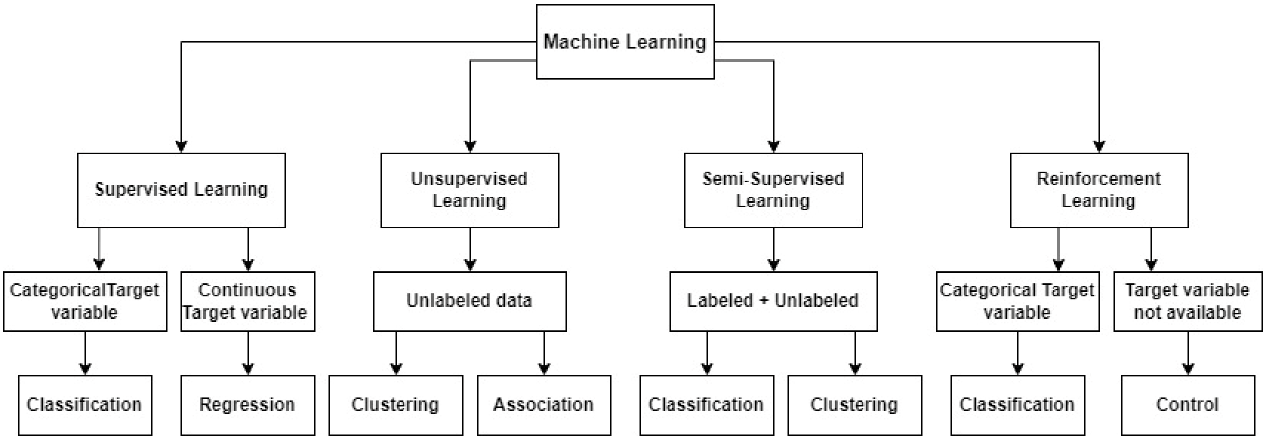

FIGURE 4. Several types of ML techniques (adapted from [16]).

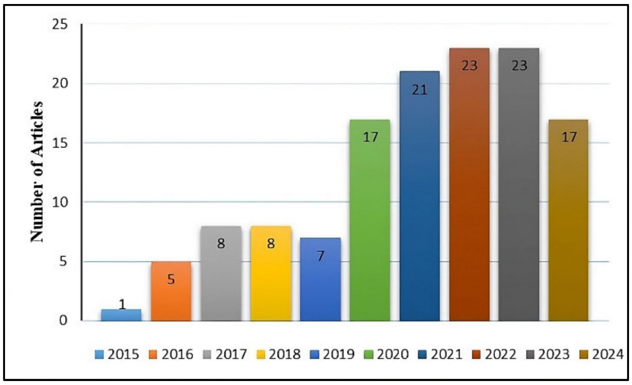

FIGURE 5. Number of included articles by year (n = 130).

in unstructured data. In contrast, semi-supervised learning combines both labeled and unlabeled data to achieve better performance in a given task [20], [21]. By leveraging the additional information from unlabeled data, the model can learn more effectively and efficiently. This approach is particularly useful in cases in which labeled data are scarce or expensive to obtain, but abundant unlabeled data are available.

4) REINFORCEMENT LEARNING

Reinforcement learning is a subfield of ML that focuses on how agents or systems can learn to make decisions and optimize their behavior, according to the concept of rewards and punishments in an environment [22], [23]. Reinforcement learning strongly emphasizes learning by an agent through direct contact with its environment, in contrast to other learning techniques such as supervised learning and unsupervised learning [24]. This method is particularly useful when the desired behavior of the agent is not well specified, or when too much variability or uncertainty exists in the environment to be captured by a fixed set of rules. Reinforcement learning is suitable for solving problems involving sequential decision-making in dynamic environments. This method is commonly used in robotics, game playing, autonomous driving, and resource allocation problems [25]. Overall, the applicability of each learning technique depends on the nature of the problem, the availability of labeled data, and the desired outcome.

IV. KEY FINDINGS REGARDING APPLICATIONS OF ML ALGORITHMS IN ERGONOMICS AND HUMAN FACTORS

A significant rise in relevant publications, especially after 2019, indicates a growing interest in the application of machine learning to HFE (Figure 5). The number of articles published in 2020 was more than twice the number published in 2019. Consequently, ML methods are gaining prominence as a solution to numerous areas in the HFE domain. Notably, 46 of the published articles focused on ANNs and DL applications in HFE.

The distribution of different types of ML algorithms during the past ten years was as follows: 46 (36%) studies applied ANNs and DL, 29 studies applied SVM (22%), 24 (19%) studies applied RF, 9 (7%) studies applied KNN, 7 (5%) studies applied boosting algorithms, 7 (5%) studies applied DTs, 5 (4%) studies applied NB, and three (2%) studies applied k-means clustering algorithms (Figure 6).

The distribution allocation of ML algorithms in various HFE application areas is presented in Table 2. ML algorithms have been applied to 1) work-related musculoskeletal disorders (WRMSDs) and human postures; 2) assessment of mental workload (MWL); 3) physical workload (PWL); 4) occupational safety and health; 5) manual material handling; 6) human factors (HF); 7) environmental ergonomics; and 8) anthropometric measurements. The distribution of the primary application domains among studies was as follows: 51 studies focused on WRMSDs and human postures (39%), 46 (36%) studies assessed MWL, 13 (10%) studies evaluated PWL, 8 (6%) studies focused on manual material handling, 4 (3%) studies addressed HF, 3 (2%) studies considered issues related to environmental ergonomics, 3 (2%) studies evaluated anthropometric measurements, and 2 (2%) study focused on occupational health and safety (Figure 7).

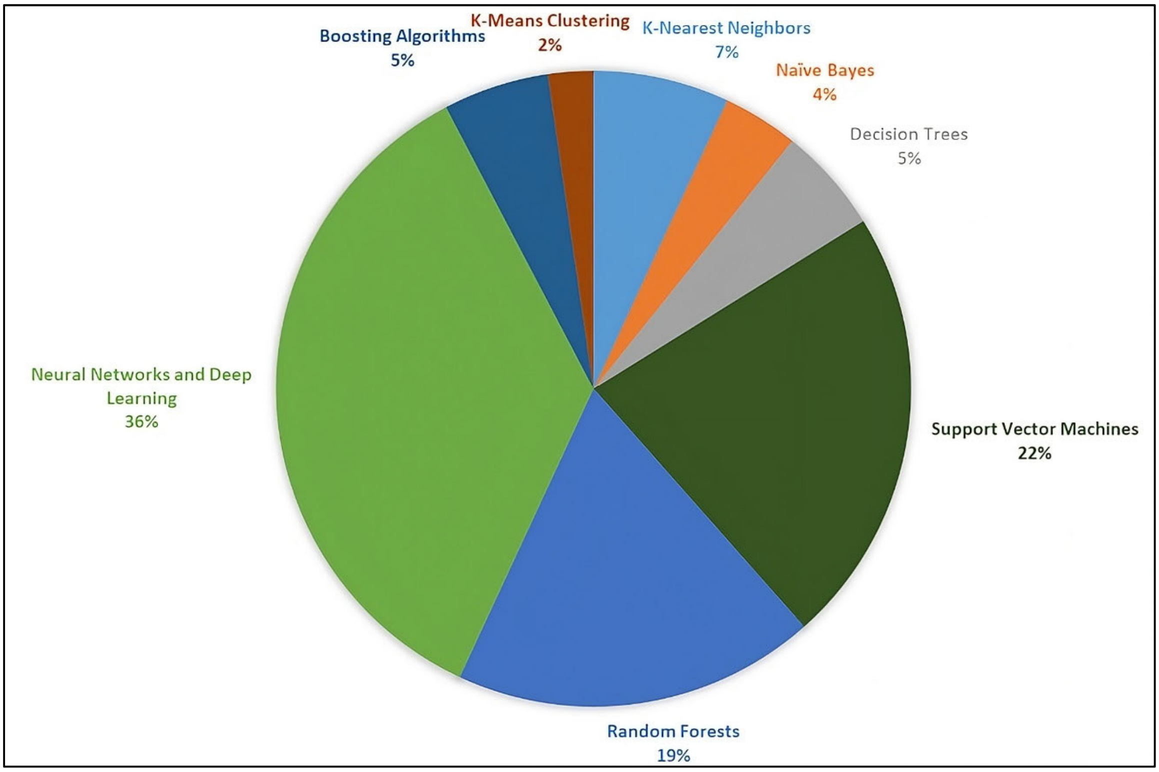

FIGURE 6. Distribution of various categories of ML algorithms over the past decade.

The selected articles were published in 70 different journals. Sensors (MDPI) published the most articles (n = 16), followed by Applied Sciences (MDPI) (n = 8), Applied Ergonomics (Elsevier) (n = 5), International Journal of Industrial Ergonomics (Elsevier) (n = 5), and Advanced Engineering Informatics (Elsevier) (n = 4). Other 65 journals published at least one article on ML applications in HFE between January 2015 and December 2024.

A total of 1074 keywords were used in the reviewed articles. Table 3 illustrates the top ten keywords used by researchers; "machine learning," "mental workload," "classification," "deep learning," and "posture" were the most frequent keywords appearing in the keyword section. The terms "machine learning" and "mental workload" were most frequently used to summarize the primary topic of the examined publications, appearing 37 (3.44%) and 27 (2.51%) times, respectively. A word cloud map visually represents textual data by displaying words in different sizes, colors, and orientations according to their frequency or significance within a dataset. In Figure 8, common words are depicted in the word cloud map, where the most frequently used words appear in larger fonts, while those used less often are shown in smaller fonts.

V. APPLICATIONS OF ML ALGORITHMS IN THE FIELD OF HFE

Several ML algorithms including KNN, NB, DTs, SVM, RF, ANNs, DL, boosting methods, and k-means clustering were examined to assess their effectiveness in supporting various HFE applications. The considered ML algorithms have been classified according to their best performance, as reported in the reviewed articles.

A. K-NEAREST NEIGHBORS

KNN is an instance-based or non-generalizing learning algorithm [26] which does not create a generalized model during the training phase. Instead, it memorizes the entire training dataset. The KNN algorithm uses similarity measures to identify the k nearest neighbors to a new data point in the training dataset [27]. A majority vote of the classes of the new data point's k nearest neighbors is then used to decide its class. KNN is quite robust to noisy training data and can be used for both classification and regression. One major challenge in KNN is choosing the optimal value of k, or the number of neighbors to consider in the classification or regression calculation.

Several studies have applied KNN algorithms in HFE (Table 4). Makaram et al. [28] compared the performance of binary and weighted versions of the visibility graph algorithms to differentiate non-fatigue and fatigue conditions. The results indicated that the extreme learning machine (ELM) classifier exhibited the highest accuracy (89.1%) and effectively-identified fatigue. The ELM classifier also had the highest specificity (92.3%) and F score (88.7%), indicating a lower misclassification rate in fatigue studies. KNN performed slightly better in identifying non-fatigue conditions

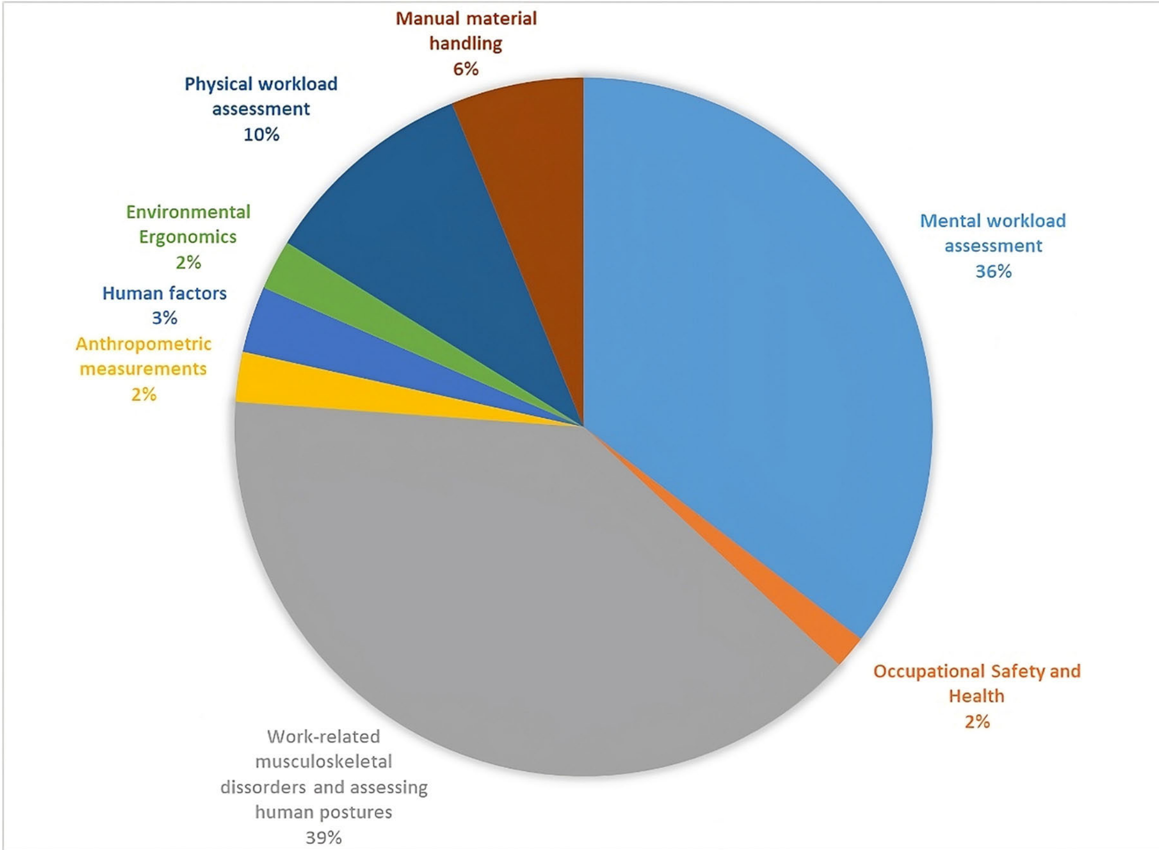

FIGURE 7. Number of articles per application area (n=130).

TABLE 2. Distribution of ML algorithms across HFE application areas.

Application area	ML algorithm							
	NN and DL	NB	DT	SVM	RF	KNN	Boosting algorithms	K-means clustering
MWLA	15	2	2	14	8	3	3	-
WRMSDs and HPA	22	1	3	10	9	2	2	1
PWA	5	1	-	-	5	2	-	-
MMH	2	-	2	2	1	-	1	-
AM	-	-	-	1	-	1	-	1
OSH	-	-	-	1	-	1	-	-
EE	1	-	-	1	-	-	1	-
HF	1	1	-	-	1	-	-	1
	46	5	7	29	24	9	7	3

*MWLA: mental workload assessment; WRMSDs and HPA: work-related musculoskeletal disorders and human posture assessment; PWLA: physical workload assessment; MMH: manual material handling; AM: anthropometric measurements; OSH: occupational safety and health; EE: environmental ergonomics; HF: human factors; KNN: k-nearest neighbors; NB: naive bayes DT: decision trees; SVM: support vector machines; RF: random forest; NN and DL: neural network and deep learning

(the k values varied between 3 and 15). Muller et al. [29] reported a foot placement classification and analysis method based on ML using a weighted KNN algorithm. The classification accuracy of the method was approximately 97% (assessed with a holdout method).

Aiello et al. [30] have demonstrated the efficacy of using a KNN classifier (considering different values of k (5, 7, and 9)) to recognize workers' actions according to attributes derived from vibration signals gathered during their operations. The KNN classifier achieved exceptionally high performance. Sánchez et al. [31] have proposed a KNN classification system (optimal number of nearest neighbors=47) to measure WRMSD risk factors, achieving an accuracy rate of 87%. Huang et al. [32] have proposed an innovative approach to evaluating the video learning effect by incorporating electroencephalography (EEG) signals and ML techniques,

TABLE 3. Most frequent keywords.

Keywords	Count	Percentage (%)
Machine learning	37	3.44
Mental workload	27	2.51
Classification	23	2.14
Deep learning	15	1.39
Posture	13	1.21
Musculoskeletal disorders	12	1.11
Ergonomics	10	0.93
EEG	10	0.93
Low back pain	7	0.65
Human factors	7	0.65

FIGURE 8. World cloud map of significant keywords.

specifically using the KNN regression algorithm for MWL testing. When the value of k is 2, the determination coefficient of the test set is 0.94. Shao et al. [33] have analyzed heart rate variability (HRV) signals as a physiological measure to assess MWL. For the classification of MWL, the study used the KNN classifier. For individual participants with a 3-minute HRV signal length, the KNN classifier achieved the best performance in classifying MWL. Similarly, Shao et al. [34] explored six classifications for MWL. They indicated that the KNN classifier (the 10-fold cross-validation was repeated 100 times) obtained the highest average recognition accuracy when used to identify six different cases separately.

Rostamzadeh et al. [35] conducted a study examining the application of different ML algorithms to identify the socio-demographic factors and hand–forearm anthropometric measurements that could reliably forecast hand function. The classifiers evaluated included k-nearest neighbor, adaptive boosting, and RF, which attained the highest classification accuracies of 96.75%, 86.49%, and 84.66%, respectively, in predicting three distinct types of pinches. Pereira and Plácido da Silva [36] developed a smart chair system incorporating posture and electrocardiography monitoring modules, leveraging an "invisible" sensing method to improve working conditions without interfering with routine activities. For posture classification, ML models were trained on datasets derived from center-of-mass coordinates on the seat plane, calculated using weight data from load cells embedded in the chair. These models achieved high classification accuracy for five sitting positions (>97.4%) and demonstrated promising results for seven positions, with the k-NN model performing the best at 87.5% accuracy.

B. NAÏVE BAYES

The NB algorithm, based on Bayes' theorem, assumes independence between each pair of features [37]. The key advantage of NB is its simplicity and speed: this method requires less training data than more complex approaches [27]. In addition, NB is relatively robust to noise in data, and it can construct a robust prediction model with the provided training data [38]. However, a major challenge in NB is its strong assumption of independence between features. In practice, many real-world datasets have features that are correlated, thus negatively affecting the performance of NB. Despite this limitation, NB remains widely used, because of its simplicity

TABLE 4. Applications of the KNN algorithm in the field of HFE (n=9).

Study	Aim	Application area, industry/sector
Aiello et al. (2022) [30]	Recognize workers' activities according to features extracted from vibration signals	Occupational safety and health, agriculture
Huang et al. (2022a) [32]	Recommend a methodology for evaluating the impact of video-based learning by utilizing EEG signal analysis in conjunction with ML algorithms.	Assessment of MWL, education
Makaram et al. (2021) [28]	Differentiate non-fatigue and fatigue conditions	Assessment of PWL, sports
Muller et al. (2019) [29]	Perform foot placement classification	Assessment of PWL, lab environment
Pereira & Plácido da Silva (2023) [36]	Present a smart chair system integrated with modules for posture monitoring and electrocardiography analysis.	WRMSDs and assessment of human postures, lab environment
Rostamzadeh et al. (2024) [35]	Evaluate the socio-demographic factors and hand-forearm anthropometric parameters that can reliably assess hand functionality	Anthropometric measurements, lab environment
Sánchez et al. (2016) [31]	Measure WRMSD risk factors	WRMSDs and assessment of human postures, general working population
Shao et al. (2021a) [33]	Assess MWL during human-robot interaction	Assessment of MWL, automotive
Shao et al. (2020) [34]	Classify operator MWL by using the HRV signal	Assessment of MWL, automotive

TABLE 5. Applications of the NB algorithm in the field of HFE (n=5).

Study	Aim	Application area, industry/sector
Ktistakis et al. (2022) [39]	Investigate the capabilities of eye and gaze data offered by the Cognitive Load Estimation Toolkit for assessing levels of cognitive workload	Assessment of MWL, lab environment
Madeira et al. (2021) [43]	Present an approach for identifying and categorizing human component categories	HF, aviation
Sun et al. (2023) [41]	Quickly predict occupant injury risks and associated uncertainties	Assessment of PWL, automotive
Thiry et al. (2022) [42]	Identify LBP conditions	WRMSDs and assessment of human postures, healthcare
Wu et al. (2021) [40]	Evaluate changes in trainees' cognitive and behavioral states during a robotic surgeon training curriculum	Assessment of MWL, healthcare

and effectiveness in many common classification scenarios. Several studies have applied NB algorithms in HFE (Table 5). Ktistakis et al. [39] investigated the use of eye and gaze data from the Cognitive Load Estimation Toolkit to assess cognitive workload levels. Among the methods tested, the Gaussian NB classifier demonstrated the highest efficiency, achieving up to 59% accuracy in predictions. Wu et al. [40] employed eye-tracking and EEG sensors to gather real-time data, concentrating on the variations that transpired between training sessions. A total of 122 observations were divided into three groups of 41, 41, and 40 for the purpose of k-fold cross-validation. Of the three classifiers assessed, the Naïve

Bayes classifier demonstrated superior performance, attaining an average accuracy of 72.2%.

Sun et al. [41] performed a Gaussian process-based surrogate model to efficiently estimate occupant injury risks and their associated uncertainties. The model achieved R^2 values of 0.9998 and 0.9991 for predictions on the left and right panels, respectively. Thiry et al. [42] assessed the effectiveness of various ML algorithms and Sample Entropy, a measure of motion variability complexity, in identifying LBP conditions. Among the tested algorithms, Gaussian NB emerged as the most effective, achieving an accuracy of 79% in identifying chronic LBP patients.

Human factors represent the leading cause of safety incidents. Over the years, advanced prediction systems have been created to assess human conditions and mitigate risks, aiming to identify and address human factors. Nevertheless, the absence of substantial and relevant labeled data has frequently hindered the progress of these systems. Madeira et al. [43] proposed a methodology for identifying and categorizing human factor categories in aviation incident reports. Based on the results, the Bayesian optimization approach demonstrated superior performance, significantly outperforming the baseline model.

C. DECISION TREES

Decision trees (DTs) are powerful models that can be used for both classification and regression tasks. They help organize a dataset into smaller, more homogeneous subsets according to different features, thus forming a tree-like structure [44]. A comparison on a particular feature is represented by each

TABLE 6. Applications of DTs in the field of HFE (n=7).

Study	Aim	Application area, industry/sector
Gómez (2020) [49]	Predict work-related musculoskeletal discomfort	WRMSDs and assessment of human postures, meat industry
Hanumegowda and Gnanasekaran (2022) [48]	Forecast important risk factors associated with WRMSDs	WRMSDs and assessment of human postures, transportation
Hussain et al. (2021) [47]	Classify driving states	Assessment of MWL, transportation
Kim et al. (2023) [51]	Forecast responses to three questions regarding the intention to use Occupational exoskeletons	WRMSDs and assessment of human postures, construction
Matijevich et al. (2021) [46]	Develop an accurate automated method for monitoring time series lumbar moments during various material handling tasks	Manual material handling, lab environment
Mudiyanselage et al. (2021) [45]	Identify body movements automatically to prevent potential muscle injuries during material handling	Manual material handling, lab environment
Su et al. (2023) [50]	Analyze risk patterns for WMSDs among sewing machine operators	WRMSDs and assessment of human postures, manufacturing

node in the tree, and the result of that comparison is represented by each branch. After navigation through the tree, the final judgment or forecast is represented by the leaf nodes. In some HFE applications, DTs have superior characteristics to other ML algorithms in classification and regression tasks (Table 6). Mudiyanselage et al. [45] have evaluated the effectiveness of surface electromyogram (EMG)–based systems combined with ML algorithms to automatically detect body movements associated with manual material handling. Based on the NIOSH lifting equation, four distinct ML models—DT, SVM, KNN, and RF—were developed to categorize risk evaluations. The DT model (using 10-fold cross-validation) achieved the highest accuracy rate. Matijevich et al. [46] constructed a precise automated technique for tracking time series lumbar moments throughout a wide range of material handling operations by merging sensor inputs with a Gradient Boosted DT algorithm. This method requires relatively few sensors while still achieving high accuracy (as high as $r^2 = 0.89$).

Hussain et al. [47] have quantified neurological biomarkers during different driving states in a virtual environment. The researchers used the C5.0 model (using 10-fold cross-validation) as an ML algorithm to classify the driving states. The C5.0 model achieved 75.38% accuracy in classifying city-roadway and expressway driving states. Hanumegowda and Gnanasekaran [48] applied ML techniques such as DT, RF, and NB to forecast important risk factors associated with WRMSDs. The DT and RF algorithms achieved high accuracy rates in predicting the relevant risk factors. Gómez [49] developed statistical models to predict work-related musculoskeletal discomfort using data from 174 workers in the meat processing industry. Logistic regression and decision tree-based models proved most effective, achieving precision rates between 83.3% and 90.2% for predicting discomfort in the shoulders, back, hands/wrists, and neck. Similarly, Su et al. [50] utilized a ML methodology to mitigate the subjectivity bias inherent in observational ergonomic evaluations and to discern risk patterns associated with WRMSDs among operators of sewing machines. They developed a decision tree comprising 13 decision rules, which resulted in a prediction accuracy exceeding 95% and a path coverage of 83%, thereby effectively illuminating risk tendencies in working postures. To comprehend the factors that influence the intention to use occupational exoskeletons and aid in making future decisions, Kim et al. [51] performed decision trees (with max_depth, min_n, and cp set to respective ranges of 2 - 10, 5 - 25, and 0.0001−0.9) to forecast responses to three questions regarding the intention to use occupational exoskeletons. The applied DT algorithm achieved a 73% accuracy rate.

D. SUPPORT VECTOR MACHINES

SVMs are based on a foundation of statistical learning theory [52]. These binary classifiers build a linear separation hyperplane to identify data instances [53]. Through the "kernel trick" to convert the original feature space into a higher-dimensional feature space, SVMs' classification skills can be greatly enhanced to enable handling of more complex data patterns. Beyond classification, SVMs can be used for regression and clustering tasks. SVMs are known for their ability to address the overfitting problems that commonly occur in high-dimensional spaces [54]. Therefore, SVMs are suitable for various applications in HFE (Table 7). For instance, Raufi and Longo [55] evaluated and analyzed the effects of the alpha-to-theta and theta-to-alpha band ratios on the development of models that can distinguish self-reported perceptions of MWL. In order to train the model, 70% of the 48 subjects were chosen randomly from both the "suboptimal MWL" and the "super optimal MWL" classes. The remaining 30% was reserved for testing the model. SVM and logistic regression were effective in discriminating among different levels of MWL according to the alpha-to-theta and theta-to-alpha band ratios.

Conforti et al. [56] reported that the SVM algorithm trained on the kinematic parameters automatically identified two distinct human body postures, achieving the highest accuracy rate. The features underwent normalization by deducting the mean and dividing by the standard deviation of each individual feature prior to the training and evaluation of the SVM classifier. The classifier's performance was then measured using a leave-one-out cross-validation technique.

The most accurate method of categorizing a seated person's posture under changing task performance conditions was reported by Roh et al. [57], who applied SVM with the radial basis function kernel (average and maximum classification rates of 97.20% and 97.94%, respectively). Nath et al. [58] used SVM to categorize risk levels brought on by physical overexertion: 1- lift/lower/carry, 2- push/pull), and 0 - other no-risk activities, with a classification accuracy of 90.2%. Luo et al. [59] performed risk prediction models specifically targeting musculoskeletal disorders of the neck and shoulder in healthcare professionals. The SVM model exhibited superior performance compared to other models, as evidenced by its lower Mean Absolute Error (MAE) and Root Mean Square Error, indicating a more stable and dependable performance when tested on previously unseen data.

Chen et al. [60] used autobiographical recollections of happy and sad emotions and a high-level MWL task to induce three distinct gait conditions: happy, sad, and MWL. An SVM classifier was used for the classification task, with accuracy rates of 83.33% for the neutral gait condition, 85.72% for the happy gait condition, 85.71% for the sad gait condition, and 89.29% for the high MWL gait condition. An experimental study conducted by Mohanavelu et al. [61] in a high-fidelity flight simulator environment successfully classified pilots' cognitive workload levels at different phases of flight using various classification algorithms, including linear discriminant analysis (LDA), SVM, and KNN based on HRV and EEG features. The analysis showed that HRV features with PCA performed better in classifying the takeoff (LDA – 75%, kNN – 60%, SVM – 75%) and landing phase (LDA – 75%, kNN – 60%, SVM – 75%). Delva et al. [62] have investigated the effects of user anthropometry on the quality of force myography (FMG) signal acquisition and the performance of ML models in classifying different hand and wrist gestures. The performance of ML models was affected by the use of raw versus normalized FMG data. Specifically, when training the extreme learning machine (ELM) model, using raw FMG data resulted in higher classification accuracy. In contrast, for the SVM, LDA, and ANN models, using normalized FMG data led to higher classification accuracy.

Mao et al. [63] introduced an ML framework for collecting data and explored the relationships among factors to develop a scalable comfort model. The SVR-RBF algorithm outperformed the other models using metrics such as the mean squared error, MAE, and R-squared score (0.36, 0.47 and 0.48 respectively). The application of ML algorithms to distinguish among several biomechanical risk categories based on electromyographic data from the back muscles has been examined by Donisi et al. [64]. The SVM method outperformed the other ML algorithms in the classification test. Similar studies have focused on classifying risk on LBP [65], [66], [67], [68], [69].

Aghajani et al. [70] have used an ML approach to detect the level of MWL by extracting various features from EEG, functional near-infrared spectroscopy (fNIRS), and a combination of EEG and fNIRS signals. These extracted features were considered biomarkers of MWL, and were used as input for a linear SVM in the training and test sets. The combination of EEG and fNIRS features and the SVM classifier effectively discriminated among different levels of MWL. Zhang et al. [71] introduced a nonlinear auto-regressive model with exogenous inputs (NARX)-based least-squares SVM (LSSVM) for identifying five levels of MWL. With a success rate of 88%, the NARX-based LSSVM method achieved the highest accuracy among other tested classifiers. In another study, Zhang et al. [72] introduced an adaptive SVM-based method for classifying operator MWL into discrete levels using psychophysiological measures. The adaptive ensemble selection model based on adaptive boosting SVM (AES-ABSVM) achieved a significant performance improvement (10–20%) over the binary SVM algorithm alone. Asgher et al. [73] have collected two-state MWL signals from the pre-frontal cortex region of participants' brains. These signals were then processed with a SVM model trained to classify the MWL signals for operating the robotic exoskeleton hand. A classification accuracy of 91.31% was achieved using a combination of mean-slope features. Other studies concentrated on utilizing the SVM-GA model to differentiate between three MWL conditions [74] and evaluating MWL from EEG data using SVM [75].

Kakkos et al. [76] have estimated the multi-frequency power spectrum and functional connectivity in two working-memory tasks completed by 40 healthy participants at two workload levels. The significance of each feature vector and the cross-task classification performance were then evaluated with ML techniques. Various classifiers were used to assess classification performance, such as Gaussian SVM, KNN, LDA, and RF. No classifiers outperformed the linear SVM in terms of classification accuracy, but all had good performance. Abdurrahman et al. [77] have examined how neuro-cognitive load affects learning transfer through the development of a driving system based on virtual reality. The classifier takes the preprocessed vector as input and outputs the level of cognitive load. The research used five well-known ML algorithms to classify cognitive workload: discriminant analysis, KNN, DT, ANNs, and SVM. Among these algorithms, SVM achieved the best accuracy for performance features and pupil dilation. Cardone et al. [78] have applied ML algorithms to infrared and HRV data to precisely measure the cognitive effort levels of participants driving in a simulated setting. Performance was compared among six classifier categories—DT, discriminant analysis, logistic

TABLE 7. Applications of SVM in the field of HFE (n=29).

Study	Aim	Application area, industry/sector
Abdollahi et al. (2020) [68]	Classify patients with nonspecific LBP	WRMSDs and assessment of human postures, healthcare
Abdurrahman et al. (2021) [77]	An immersive driving system utilizing virtual reality to examine the impact of neuro-cognitive load on the transfer of learning.	Assessment of MWL, transportation
Aghajani et al. (2017) [70]	Detect the level of MWL	Assessment of MWL, lab environment
Antwi-Afari et al. (2018) [69]	Identify and categorize awkward working postures.	WRMSDs and assessment of human postures, construction
Asgher et al. (2021) [73]	Classify the MWL signals for operating a robotic exoskeleton hand	Assessment of MWL, lab environment
Baghdadi et al. (2018) [81]	Classify non-fatigued vs. fatigued states following manual material handling	Manual material handling, lab environment
Cao et al. (2022) [79]	Classify MWL	Assessment of MWL, lab environment
Cardone et al. (2022) [78]	Estimate the cognitive workload levels of participants driving in a simulated environment	Assessment of MWL, transportation
Chen et al. (2023) [60]	Incorporate autobiographical narratives that express both joyful and sad emotions, together with a high MWL task, to facilitate the emergence of three gait conditions: happiness, sadness, and MWL.	Assessment of MWL, lab environment
Conforti et al. (2020) [56]	Automatically identify two postures with high accuracy	WRMSDs and assessment of human postures, wearable technology
Dandumahanti & Subramaniyam (2023) [66]	Differentiate between neutral and flexed neck postures	WRMSDs and assessment of human postures, lab environment
Delva et al. (2020) [62]	Investigate the effects of user anthropometry on the quality of FMG signal acquisition	Anthropometric measurements, wearable technology
Donisi et al. (2023) [64]	Discriminate among biomechanical risk classes according to EMG signals of back muscles	Manual material handling, wearable technology
Kakkos et al. (2021) [76]	Assess the multi-frequency power spectrum and functional connectivity across two workload levels in two distinct working-memory tasks.	Assessment of MWL, lab environment
Lamichhane et al. (2021) [67]	Distinguish LBP patients from healthy controls	WRMSDs and assessment of human postures, healthcare
Lee at al. (2021) [82]	Implement an automated method for evaluating the perceived risk levels among construction workers	Occupational safety and health, construction
Luo et al. (2024) [59]	Develop predictive models to evaluate the probability of musculoskeletal disorders affecting the neck and shoulders in healthcare professionals.	WRMSDs and assessment of human postures, healthcare
Mao et al. (2019)[63]	Develop a scalable comfort model	Environmental ergonomics, office environment
Matos et al. (2024) [65]	Prevent WRMSDs in manufacturing industries	WRMSDs and assessment of human postures, manufacturing
Meteier et al. (2021) [80]	Classify the workload of drivers in real time	Assessment of MWL, transportation
Mohanavelu et al. (2022) [61]	Classify pilots' cognitive workload in a high-fidelity flight simulator	Assessment of MWL, aviation
Nath et al. (2018) [58]	Evaluate ergonomic risk levels caused by overexertion	WRMSDs and assessment of human postures, wearable technology
Raufi and Longo (2022) [55]	Discriminate among levels of MWL according to the alpha-to-theta and theta-to-alpha band ratios	Assessment of MWL, lab environment
Roh et al. (2018) [57]	Identify the most accurate method for classifying the actual sitting posture of a seated person	WRMSDs and assessment of human postures, office environment
Safari et al. (2024) [75]	Suggest innovative approaches for evaluating MWL using EEG data, leveraging effective brain connectivity for feature extraction	Assessment of MWL, lab environment
Seo and Lee (2021) [83]	Proposed a vision-based approach for the automatic classification of workers' postures to evaluate ergonomics using SVM.	WRMSDs and assessment of human postures, construction
Wu et al. (2017) [74]	Collect handwriting data across different MWL conditions	Assessment of MWL, lab environment
Zhang et al. (2017b) [71]	Identify five levels of MWL	Assessment of MWL, lab environment
Zhang et al. (2014) [72]	An adaptive SVM-based method for classifying operator MWL	Assessment of MWL, lab environment

regression, SVM, nearest neighbor, and ensemble classifiers. A k-fold cross-validation with k set to 5 was utilized to prevent overfitting. The best-performing models applied a linear kernel and two classes of SVM classifiers.

In the study by Cao et al. [79], an SVM algorithm was used for MWL classification. The SVM algorithm constructed an optimal hyperplane in the feature space by minimizing the structural risk, thus enabling accurate workload classification. The fNIRS and EEG data were used to extract the selected features, which were then input into an SVM by using a radial basis function kernel. The researchers also tested and compared ML algorithms including KNN, DT, and LDA and found that the SVM algorithm outperformed the other methods in MWL classification. Meteier et al. [80] have evaluated and classified the workload of drivers in real time by using ML techniques. The authors selected three algorithms for this task: RF classifier, C-support vector classifier, and multi-layer perceptron classifier. The C-support vector classifier model performed best, achieving an accuracy of 95% in classifying the condition of the driver's workload. Baghdadi et al. [81] conducted a study where they utilized a SVM model to differentiate between non-fatigued and fatigued states during manual material handling. The classification process involved setting aside 20% of the feature data points for testing while the SVM was trained using the remaining data points. A 5-fold cross-validation was performed on each subject's data, with the 50 feature data points being randomly divided into five subgroups of equal sizes to validate the classifier. The categorization utilizing foot acceleration and position trajectories resulted in a 90% precision.

In their study, Lee et al. [82] introduced an automated technique for assessing the perceived risk levels of construction workers. This technique involves analyzing physiological data collected from wristband-type wearable biosensors in combination with a supervised learning algorithm. The dataset was divided into ten subsets of equal size during the 10-fold cross-validation process. Nine subsets were used for training the model, while one subset was reserved for evaluating the model's performance. This training and validation procedure was iterated ten times to ensure each subset was used for validation purposes. The Gaussian SVM algorithm demonstrated superior performance compared to other algorithms. Seo and Lee [83] proposed a vision-based approach for automatically classifying workers' postures to evaluate ergonomics using SVM. The results of the experiments indicated an approximately 89% accuracy rate in automatically classifying various postures in images, validating the efficacy of virtual training images for posture classification.

E. RANDOM FOREST

The RF algorithm [84], is an ensemble learning method within the broader category of DT-based models. The RF model uses a technique called bootstrap aggregating or bagging. Multiple subsets of the original dataset are created by sampling with replacement (bootstrap samples). Each subset is then used to train a separate regression tree. The key distinction of RF is in the independence of the training process for each tree. Because each tree is trained on a different bootstrap sample, the trees are decorrelated, thus preventing overfitting and improving the robustness of the model. Key components of RF algorithms involve the number of trees, the minimum observations needed in a terminal node, and the number of appropriate features for splitting [85], [86]. RF is known for its versatility, robustness, and ability to handle high-dimensional data [87]. RF is widely used in applications including classification, regression, and feature importance analysis.

Several authors have proposed RF algorithms in HFE (Table 8). Duan et al. [88] introduced an automated and non-invasive approach for identifying the material handling tasks of construction workers during the pandemic, employing smartphones and ML techniques. The classification performance for the shoulder handling technique slightly surpassed that of the anterior handling technique. Among the five classifiers evaluated, cross-validation results demonstrated that the RF algorithm attained the highest classification accuracy, recording 76.71% for the anterior handling technique and 80.13% for the shoulder handling technique, utilizing a window size of 0.64 seconds. Markova et al. [89] investigated the correlation between LBP and extended periods of inadequate sitting posture by utilizing photogrammetric imaging, calculating postural angles, employing machine learning models, and administering self-reported questionnaires regarding LBP and associated symptoms among the study participants. The random forest classifier surpassed other algorithms, attaining an accuracy of 80%, a recall rate of 93.3%, and a precision of 79.0%, thereby showcasing strong and reliable performance. Ferrone et al. [90] have investigated workers' perceptions of LBP and their willingness to use wearable monitoring technologies to decrease injury risk. The collected data were analyzed with the RF model resulting in an accuracy of 91.66%. Similar studies have focused on assessing risk on LBP [91], [92], [93], [94], [95] and sitting position [96].

Nogueira et al. [97] have proposed a predictive model to forecast fatal occurrences in aviation events, including accidents and incidents. Three supervised learning algorithms were considered: RF, ANNs, and the semi-supervised active learning approach. The RF algorithm achieved the best results, with an accuracy of 90%. Villalobos and Mac Cawley [98] have discussed the automation of artificial intelligence and ML techniques for task classification and ergonomic assessments in work environments, as well as the use of inertial measurement units to evaluate human activity. The RF algorithm for classifying the RULA scores for ergonomic assessments of the wrist/hand achieved the highest accuracy score, along with the highest precision, recall, and F1 scores. Manjarres et al. [99] have used ML algorithms to incorporate the concept of Frimat's score from ergonomics to calculate the corresponding PWL according to measured heart rate values. In real-time tests with

20 participants, the RF activity classifier achieved an accuracy of 92%. Alternatively, some research has been dedicated to identifying and classifying the degrees of physical fatigue among construction workers [100], [101]. To reduce subjectivity in overall discomfort rating (ODR) responses, Hota et al. [102] ML algorithms such as KNN, RF classifier, and SVM were used to predict ODR based on body mass index, heart rate, estimated energy requirement, and electromyography. These models achieved high prediction accuracies ranging from 87% to 97%, with RFC demonstrating the highest accuracy.

Park et al. [103] have investigated the effectiveness of classification models in assessing cognitive workload in EMG-based prosthetic devices, for both choice reaction time and shape pattern tasks. The RF and NB algorithms achieved greater accuracy than random guessing (i.e., 0.5) in all 15 runs when the target variable (i.e., cognitive workload) was classified into two clusters. Wei et al. [104] have analyzed the MWL of drivers by considering multiple factors such as physiological signals, traffic flow, and environmental conditions. The classification performance of single-factor and multi-factor data by using three ML algorithms: ANNs, SVM, and RF was assessed. Notably, the RF algorithm achieved outstanding classification results with 97.8% accuracy. Badarna et al. [105] have proposed the concept of pen motion pattern groups and investigated their role in classifying handwriting, on the basis of cognitive MWL. To support their claim, the authors developed a semi-automatic ML-based classification framework using an RF classifier. In a study by Borghetti et al. [106], an RF supervised ML algorithm have been applied to EEG input to infer operator workload by using Improved Performance Research Integration Tool workload model estimates. The study utilized the *Leave One Out Cross Validation* method to assess models trained on all participants except one. This approach involved assigning each participant's data to a fold and testing on the last one. In another study by Lin and Lukodono [107], human MWL is assessed quantitatively in Human–Robot Collaboration through physiological feedback gathered via wearable sensors. This feedback is analyzed to recognize human workload perception using a classification approach with supervised ML algorithms. The results demonstrate that the RF algorithm, which selects features based on information gain, achieves an accuracy of up to 94% in classifying MWL during the testing phase.

Le et al. [108] have sought to determine whether supervised learning could be used in conjunction with near-infrared spectroscopy data obtained from an actual driving environment to categorize drivers' mental effort. The RF model outperformed other algorithms and achieved the highest accuracy in this classification task. Islam et al. [109] have introduced a methods combining EEG and vehicular signals to construct a mutual information-based feature set. This feature set was then used to evaluate drivers' MWL in a real driving experiment. The training and validation ratios were set at 90% and 10%, respectively. The experiment was repeated ten times, and the average outcomes were displayed. The RF classifier, using mutual information-based features, demonstrated the best performance among all methods evaluated in this study, with a high accuracy rate of 94%. To further examine the effects of sensor fusion and task performance on the physiological state of automated vehicle drivers, Meteier et al. [110] have used ML techniques to perform classification and regression tasks. For the three-level classification task, MWL was predicted with an accuracy of 71% (measured with the F1 score) by using the electrodermal activity and respiration signals as inputs to an RF classifier. In their study, Maman et al. [111] forecasted alterations in employee productivity to inform recommendations for rest breaks, job rotations, and task allocations. The RF model achieved the highest prediction performance.

F. ANNS AND DL

ANNs are basic mathematical models inspired by the brain's functioning. They process information and generate predictions by performing non-linear mappings or detecting patterns in data. ANNs are especially effective in modeling complex and non-linear structures [112], [113], [114]. At the core of an ANN is the neuron, which serves as the fundamental processing unit. Neurons use transfer functions that formulate their output according to the input received. A major advantage of ANN models is their ability to handle multiple-variable problems with lower complexity rules. Instead of explicitly defining complex rules, ANNs can learn patterns and extract crucial information within complex information domains. Several studies applying ANN algorithms in HFE have been reported (Table 9). For instance, Jiang et al. [115] have enhanced the classification of complex EEG data by using a long short-term memory (LSTM) ANN model with an attention mechanism. The final model achieved a 94% detection accuracy for 2-second EEG data.

To improve the flexibility of robotic elements, Oliff et al. [116] have examined human variables and proposed a system combining data processing and intelligent control. ANN was used to predict the effects on human task performance. A continuous variable was scaled to fit within the range of -1 to 1 in order to avoid the saturation of the sigmoid activation functions. Categorical variables were transformed using a "one-hot" encoding method, where each category is represented as a series of binary input nodes, resulting in a total of 11 input nodes. These predictions were then used to make informed adjustments to the programmed behavior of the robots. Shao et al. [117] have used bidirectional long short-term memory (Bi-LSTM) along with several ML algorithms to classify the levels of MWL among participants. The Bi-LSTM classification approach performed best when used with both the suggested features and the original EEG data. The average accuracy and specificity (85.61% and 89.64%, respectively) were highest when Bi-LSTM was used with the original EEG data plus FGMSM characteristics. Similarly, a classification model able to identify various physical strain levels has been developed by Yang et al. [118]. Bi-LSTM,

TABLE 8. **Applications of RF algorithms in the field of HFE (n=24).**

Study	Aim	Application area, industry/sector
A. Abdel Hady and Abd El-Hafeez (2024) [91]	Examine trunk movement patterns in women experiencing postpartum LBP	WRMSDs and assessment of human postures, lab environment
Antwi-Afari et al. (2023) [101]	Detect and classify physical fatigue levels in construction workers	Assessment of PWL, construction
Anwer et al. (2023) [100]	Identify and categorize physical fatigue among construction workers by analyzing heart rate variability	Assessment of PWL, construction
Badarna et al. (2018) [105]	Propose the concept of pen motion pattern groups and investigate their role in classifying handwriting, on the basis of cognitive MWL	Assessment of MWL, lab environment
Borghetti et al. (2017) [106]	Apply RF algorithm to EEG input to infer operator workload	Assessment of MWL, lab environment
Duan et al. (2023) [88]	Identify material handling tasks performed by construction workers using ML techniques	Manual material handling, construction
Ferrone et al. (2021) [90]	Investigate workers' perceptions of LBP and their willingness to use wearable monitoring technologies to decrease injury risks	WRMSDs and assessment of human postures, healthcare
Hota et al. (2023) [102]	Predict overall discomfort rating based on body mass index, heart rate, estimated energy requirement, and electromyography	Assessment of PWL, lab agriculture
Islam et al. (2020) [109]	Evaluate drivers' MWL in a real driving experiment	Assessment of MWL, transportation
Kar et al. (2023) [92]	Identify the risk factors contributing to WRMSDs among dumper operators in Indian iron ore mines	Assessment of MWL, mining
Le et al. (2018) [108]	Investigate the feasibility of classifying driver MWL with near-infrared spectroscopy data collected from a real vehicle setting	Assessment of MWL, transportation
Lin & Lukodono (2022) [107]	Classify of MWL in human-robot collaboration	Assessment of MWL, lab environment
Maman et al. (2020) [111]	Predict changes in workers' performance	Assessment of PWL, lab environment
Manjarres et al. (2019) [99]	Calculate the corresponding PWL based on measured heart rate values	Assessment of PWL, wearable technology
Markova et al. (2024) [89]	Investigate the relationship between LBP and extended periods of inadequate sitting posture.	WRMSDs and assessment of human postures, lab environment
Meteier et al. (2022) [110]	Predict and classify MWL among vehicle drivers	Assessment of MWL, transportation
Nogueira et al. (2023) [97]	Predict fatal occurrences in aviation events, including accidents and incidents	HF, aviation
Park et al. (2023) [103]	Examine the effectiveness of classification models in evaluating cognitive workload in EMG-based prosthetic devices.	Assessment of MWL, lab environment
Sasikumar and Binoosh (2020) [94]	Predict the likelihood of musculoskeletal disorders in computer professionals	WRMSDs and assessment of human postures, office environment
Tomkins-Lane et al. (2022) [93]	Analyze physical activity features linked to LBP using real-world accelerometry data	WRMSDs and assessment of human postures, healthcare
Villalobos and Mac Cawley (2022) [98]	Conduct task classification and ergonomic assessments in workplace environments.	WRMSDs and assessment of human postures, meat industry
Wei et al. (2023) [104]	Analyze the MWL of drivers by considering multiple factors such as physiological signals, traffic flow, and environmental conditions	Assessment of MWL, transportation
Zemp et al. (2016a) [95]	Investigate how back pain affects sitting behavior in an office setting.	WRMSDs and assessment of human postures, office environment
Zemp et al. (2016b) [96]	Identify the user's sitting position	WRMSDs and assessment of human postures, office environment

a DL technique renowned for its exceptional performance in time series signal estimation, was used for this purpose. In the binary classification between circumstances of no load and any load, Bi-LSTM achieved a high accuracy. Zhang et al. [119] have addressed the MWL classification problem by using measured physiological data. The proposed ensemble convolutional neural network framework effectively improved MWL classification performance. Similar studies have focused on developing MWL prediction models [120], [121], [122], [123], [124], [125].

A semi-supervised extreme learning machine (SS-ELM) technique has been presented by Zhang et al. [126] for the categorization of MWL patterns. The research compared the SS-ELM technique with four standard supervised learning algorithms—NB, RF, SVM, and ELM—to assess its potential for MWL classification. In comparison to the four main supervised learning algorithms, the SS-ELM approach achieved greater classification accuracy by using a large amount of unlabeled data. In another study, Tao et al. [127] have used an extreme learning machine, a fast-training ANN, as the foundation for constructing an individual-specific classifier ensemble to recognize binary MWL. They compared the performance of their proposed approach, called HE-ELM, against that of several other algorithms, including NB, ANN, KNN, denoising autoencoder, and logistic regression. HE-ELM outperformed the other algorithms and achieved the highest performance in recognizing binary MWL. Teoh Yi Zhe and Keikhosrokiani [128] have used particle swarm

TABLE 9. ANN and DL applications in the field of HFE (n=46).

Study	Aim	Application area, industry/sector
Aghamohammadi et al. (2024) [143]	Evaluate ergonomic risks under free-occlusion and self-occlusion scenarios	Assessment of PWL, sports
Aguilar et al. (2024) [161]	Develop individualized models for vehicle drivers exposed to whole-body vibration	Assessment of PWL, construction
Amalia et al. (2023) [159]	Create an ANN model to forecast labor rest intervals.	Environmental ergonomics, manufacturing
Antwi-Afari et al. (2022) [156]	Automatically identify and categorize various types of awkward working postures.	WRMSDs and assessment of human postures, construction
Asgher et al. (2020) [142]	Evaluate and classify four-state levels of MWL by using fNIRS hemodynamics signals	Assessment of MWL, lab environment
Darvishi et al. (2017) [135]	Predict LBP among workers in industrial settings	WRMSDs and assessment of human postures, manufacturing
Dolmans et al. (2021) [122]	Classify perceived MWL using a deep neural network	Assessment of MWL, lab environment
Donati et al. (2023) [160]	Identify workers' stress and fatigue in real time	Assessment of PWL, manufacturing
Fu et al. (2022) [150]	Evaluate the physical comfort while sitting position in an office chair	Assessment of PWL, office environment
Gholipour & Arjmand (2016) [136]	Predict the three-dimensional posture of the spine	WRMSDs and assessment of human postures, lab environment
Hartley et al. (2024) [154]	Classify non-specific LBP	WRMSDs and assessment of human postures, healthcare
Hossain et al. (2023) [137]	Predict REBA risk levels and assess full-body postural risk	WRMSDs and assessment of human postures, lab environment
Hu & Hu (2024) [124]	Perform a DL model aimed at classifying MWL based on EEG signals.	Assessment of MWL, transportation
Hu et al. (2018) [138]	Identify individuals with chronic LBP using deep neural networks.	WRMSDs and assessment of human postures, lab environment
Huang et al. (2022b) [141]	Assess an MWL wherein the authors collected various physiological signals from drivers under different driving conditions	Assessment of MWL, transportation
Jain et al. (2019) [131]	Predict pulling force values	Manual material handling, lab environment
Jiang et al. (2022) [115]	Classify complex EEG data	Assessment of MWL, aviation
Jiao et al. (2024) [157]	DL based rapid entire body risk assessment for prevention of WRMSDs	WRMSDs and assessment of human postures, lab environment
Kapse et al. (2024) [134]	Detect posture in agricultural ergonomics	WRMSDs and assessment of human postures, agriculture
Kim et al. (2018) [145]	Classify children's sitting postures	WRMSDs and assessment of human postures, lab environment
Kim et al. (2019) [146]	Classify children's sitting postures	WRMSDs and assessment of human postures, lab environment
Kutafina et al. (2021) [120]	Evaluate the feasibility of using a mobile EEG setup to track MWL	Assessment of MWL, lab environment
La Delfa and Potvin (2016) [132]	Predict manual arm strength	Manual material handling, lab environment
Lee at al. (2022) [147]	Classify nine typical sitting postures of children	WRMSDs and assessment of human postures, lab environment
Li et al. (2020) [153]	Evaluate postural risk factors associated with musculoskeletal disorders	WRMSDs and assessment of human postures, lab environment
Longo (2022) [125]	Utilize EEG data and DL methodologies to assess MWL	Assessment of MWL, lab environment
MassirisFernández et al. (2020) [5]	Design a novel approach for ergonomic risk assessment that automates the calculation of RULA scores	WRMSDs and assessment of human postures, lab environment
Muşat et al. (2022) [129]	Categorize postures and differentiate between dynamic and static back work.	WRMSDs and assessment of human postures, lab environment
Ogundokun et al. (2022) [144]	Recognize and classify different postures	WRMSDs and assessment of human postures, lab environment
Oliff et al. (2020) [116]	Investigate HF and propose a framework integrating intelligent control and data processing to enhance the adaptability of robotic elements	HF, automation
Piñero-Fuentes et al. (2021) [152]	Utilize postural detection for workers	WRMSDs and assessment of human postures, manufacturing
Pušica et al. (2024) [148]	Classify MWL and identify tasks during multitasking scenarios	Assessment of MWL, lab environment

TABLE 9. *(Continued.)* **ANN and DL applications in the field of HFE (n=46).**

Shao et al. (2021b) [117]	Classify the levels of cross-subject MWL	Assessment of MWL, lab environment
Tao et al. (2019) [127]	Recognize binary MWL	Assessment of MWL, lab environment
Teoh Yi Zhe & Keikhosrokiani (2021) [128]	Predict the MWL of knowledge workers	Assessment of MWL, lab environment
Wu et al. (2020) [121]	Develop a MWL prediction model using data from non-experts	Assessment of MWL, energy sector
Xiahou et al. (2023) [139]	Identify and categorize uncomfortable working positions	WRMSDs and assessment of human postures, construction
Yan et al. (2021) [151]	Predict occupational physical activities	Assessment of PWL, lab environment
Yang et al. (2020) [118]	Develop a classification model to identify various levels of physical load.	WRMSDs and assessment of human postures, construction
Zhang & Huang (2024) [158]	Assess WRMSDs for sanitation workers	WRMSDs and assessment of human postures, manufacturing
Zhang et al. (2017a) [119]	Address the MWL classification problem by using measured physiological data	Assessment of MWL, lab environment
Zhang et al. (2020) [126]	MWL pattern classification	Assessment of MWL, lab environment
Zhang and Li (2017) [123]	Address the MWL classification problem	Assessment of MWL, lab environment
Zhao & Obonyo (2020) [149]	Recognize construction workers' postures based on motion data	WRMSDs and assessment of human postures, construction
Zhao and Obonyo (2021) [130]	Examine the natural postures of real construction workers while they perform their daily tasks.	WRMSDs and assessment of human postures, construction
Zokaei et al. (2024) [133]	Examine and forecast the complex interactions among risk factors that lead to musculoskeletal disorders in computer users within government workplaces	WRMSDs and assessment of human postures, office environment

optimization with a microgenetic algorithm to improve the extreme learning adaptive neuro-fuzzy inference system. This improved model was used to forecast knowledge workers' mental effort. Compared with the extreme learning adaptive neuro-fuzzy inference system model optimized with particle swarm optimization alone, the proposed model achieved even greater enhancements.

Through the use of ML, Muşat et al. [129] have evaluated the viability of classifying postures and differentiating between static and dynamic back work by using triaxial acceleration data. Two ML methods, an RF and a multilayer perceptron with back propagation, were then trained on the filtered dataset. The multilayer perceptron with back propagation performed better in distinguishing between dynamic and static labor. Zhao and Obonyo [130] have used the incremental convolutional long short-term memory (CLN) model to examine the natural postures of construction workers performing their everyday jobs. According to the macro F1 score, the CLN model with as many as two convolutional layers performed well in both customized (0.87) and generic (0.84) modeling situations. Jain et al. [131] conducted a comparative study to evaluate the effectiveness of ANNs in relation to regression models for predicting pulling force values. The findings indicated that ANN models exhibited a markedly greater explained variance and reduced root mean square difference values for PF data across three different handle heights. Similarly, La Delfa and Potvin [132] assessed the performance of ANN and regression models in predicting manual arm strength (MAS) using identical development and validation datasets. Data on multi-directional MAS were gathered from 95 healthy female participants at 36 hand locations within the reach envelope. The results showed that ANN predictions provided a significantly higher explained variance (90.2% compared to 66.5% for the regression model) and a considerably lower root mean square deviation of 9.3 N versus 17.2 N. Comparable studies have utilized ANNs to address musculoskeletal disorders and posture detection [5], [133], [134], [135], [136], [137], [138], [139]. More recent ANN studies examining various applications in HFE have been discussed in a recent review by Çakıt and Karwowski [140].

Huang et al. [141] have conducted a simulated driving experiment in which various physiological signals were collected from drivers under various driving conditions. Six channels' signals were gathered and utilized as the input for the CNN, with down-sampling performed on the signals to decrease the sample scale from 500 Hz to 100 Hz. The DL model, specifically a combination of CNNs and LSTM, achieved a higher accuracy rate than the other methods. Asgher et al. [142] have evaluated and classified four-state levels of MWL by using fNIRS hemodynamics signals. They used three traditional ML algorithms (SVM, KNN, and ANN) and two DL algorithms (CNN and LSTM). The results indicate that the proposed DL algorithms, including LSTM and CNN, significantly outperform traditional ML algorithms such as SVM, ANN, and k-NN in classification performance. Aghamohammadi et al. [143] introduced a visual ergonomic assessment technique that utilizes a multi-frame and multi-path CNN to evaluate ergonomic risks under free-occlusion and self-occlusion conditions. The results demonstrate that our architecture achieves competitive performance, with a recall of 0.8925, precision of 0.8743, and an F-score of 0.8837.outperformed the traditional ML algorithms (SVM, ANN, and KNN) in terms of classification. Ogundokun et al. [144] have used four models to aid in decision-making regarding posture recognition. CNN and multilayer perceptron with two DL algorithms and two transfer learning were implemented by using images from the MPII Human-Posture dataset. The AlextNet model combined with hyperparameter optimization achieved the highest performance (validation accuracy of 91.2%), to accurately recognize and classify different postures. Similar studies have focused on CNN for classifying children's sitting postures [145], [146], [147], EEG segments based on task load levels [148], recognizing construction workers' postures [149], application of sEMG-based human-computer interaction [150], predicting occupational physical activities [151], worker postural detection [152], rapid upper limb assessment [153], classifying non-specific low back pain [154].

DL is a subset of ML within the category of supervised learning, unsupervised learning, or reinforcement learning, depending on the nature of the task. The term ''deep'' refers to the use of deep NNs, which have multiple layers (deep architectures). These deep models can learn intricate hierarchical representations of data. A distinguishing feature of DL is its ability to automatically learn relevant features from raw data [155]. In some HFE applications, DL algorithms have characteristics superior to those of other ML algorithms in classification tasks. Using wearable insole sensor data and DL-based networks, Antwi-Afari et al. [156] identified and classified several forms of difficult working postures in the construction industry. In this research, the cross-entropy loss (log loss function) was employed as the cost function to measure model classification accuracy. The GRU model performed better than the other RNN-based DL models. The conventional 2D image-based approach falls short in accurately capturing the complex 3D movements and postures involved in various occupational tasks. To overcome this limitation, Jiao et al. [157] developed an enhanced deep learning-based Rapid Entire Body Assessment (REBA) method. This method processes working videos as input and automatically generates REBA scores through 3D pose reconstruction. The proposed approach achieves an average precision of 94.7% on real-world data, matching the accuracy of ergonomic experts. Zhang and Huang [158] integrated deep learning and image recognition to develop a Quick Capture Evaluation System. By analyzing body angles captured in the work environment of sanitation workers and comparing them with data from OptiTrack motion capture, the system achieved an average RMSE of 5.64 across 18 different postures and an average Spearman's rho of 0.87, demonstrating its accuracy. Additionally, when compared to scores assessed by three experts, the system achieved an average Cohen's kappa of 0.766, confirming its reliability.

Amalia et al. [159] examined the environmental workstation, labor workload, and rest periods; and developed an ANN model for predicting labor rest periods. The findings revealed that dry bulb temperature, heart rate, wet bulb temperature, and gender had a significant influence on the rest period and Wet Bulb Globe Temperature. The R^2 value exceeded 0.900, demonstrating that the ANN model effectively predicted labor rest periods based on environmental ergonomics. Donati et al. [160] have assessed worker weariness and stress in real time by using wearable sensors and applying ML algorithms. The entire dataset was divided into three parts: 75% for training, 15% for validation, and 10% for testing purposes. The developed system had excellent usability, 88.4% accuracy and an F1 score of 0.90 in identifying stress from electrocardiogram (ECG) data when a 1D convolutional NN was used. Aguilar et al. [161] introduced a methodology aimed at developing personalized models for vehicle operators subjected to whole-body vibration. This approach is intended to be readily adopted by organizations to safeguard the health of their employees in both the immediate and extended future. A measurement campaign was carried out with the participation of a professional driver, and the resulting data were employed to develop six ANNs. These networks aim to forecast the daily compressive load on the lumbar spine and evaluate exposure to whole-body vibration over both short and extended periods. The models exhibited a significant degree of accuracy, achieving R^2 values exceeding 0.90 throughout the training, cross-validation, and testing phases.

G. BOOSTING ALGORITHMS

The concept of boosting [162] originated from a study by Schapire [163] demonstrating that a weak learner can be transformed into a strong learner through boosting. Some boosting variants, such as extreme gradient boosting (XGBoost) [164], light gradient boosting [165], and categorical boosting [166], focus on increased predictive performance and speed. Boosting algorithms have shown remarkable results in HFE applications and forecasting competitions (Table 10). For instance, Harputlu Aksu and Çakıt [167] have developed ML algorithms for classifying MWL by using eye tracking data. The gradient boosting machine algorithm demonstrated the best performance, with a classification accuracy rate of 68% for 14 eye-tracking features and 65% for 27 eye-tracking features. Similarly, Harputlu Aksu et al. [168] employed a hybrid approach incorporating EEG and eye-tracking methodologies while participants engaged in n-back tasks to tackle the issue of classifying MWL through ML algorithms. Of the various methods evaluated, the LightGBM algorithm demonstrated superior performance, attaining the highest prediction accuracy of 76.59% with the utilization of 34 selected features. Teuler et al. [169] have sought to predict the workload of a controller by analyzing the controller's actions. The authors decided to partition the data randomly into training and testing datasets. Specifically, the training set encompasses 80% of the data, leaving the remaining 20% for the test set. Several ML models were tested to explore different combinations of features that accurately predicted the controller's workload, among which two algorithms, RF and the XGBoost algorithm, were selected for their accurate predictive capabilities.

Donisi et al. [170] have classified lifting tasks belonging to different lifting index classes according to the revised NIOSH lifting equation. A stratified cross-validation method was implemented to maintain consistent class proportions across different folds. Additionally, to enhance the generalizability of the proposed approach, the authors conducted a leave-one-subject-out cross-validation with six subjects for training the predictive models and one subject for testing the algorithms. Time-domain features extracted from acceleration and angular velocity signals were used as input to several ML algorithms, and GBM was found to be the best algorithm for classifying lifting tasks based on the RNLE. Chen et al. [171] have used ECG and EMG data as physiological markers to identify tiredness in miners and to fuse multi-modal feature information. The categorization of tiredness was achieved by combining multi-modal data and applying SVM, RF, and XGBoost ML models, XGBoost was found to be the most effective in fatigue identification with 89.47% recognition accuracy. Katić et al. [172] have applied ML algorithms to predict thermal comfort levels in individuals on the basis of input data from experiments. Personal comfort models using RUSBoosted trees showed the best median accuracy (0.84) for all tested approaches (a 5-fold cross-validation was performed). Nijeweme-d'Hollosy et al. [173] commenced a study focused on the development of a clinical decision support system aimed at aiding patients suffering from LBP in their self-referral to primary care services. In pursuit of this objective, the research investigated the application of supervised ML techniques. Of the various models evaluated, the boosted tree algorithm demonstrated the highest performance, delivering referral recommendations with moderate accuracy, as indicated by a Kappa value between 0.2 and 0.4.

H. K-MEANS CLUSTERING

K-means clustering is a commonly used unsupervised ML algorithm that involves dividing a dataset into groups or clusters, with the goal of ensuring that observations within the same cluster are more similar to each other than to those in other clusters. The term "k" in K-means refers to the number of clusters that the algorithm is set to create. The algorithm aims to create clusters in which the mean value (centroid) of the observations in each cluster is placed at the center. This centroid represents the average position of all data points in that cluster [174]. K-means clustering is widely used in various fields for tasks such as customer segmentation, image compression, and document categorization. Recently, Rababaah [175] has proposed k-means clustering algorithm for the classification of human postures for monitoring of older individuals and fall detection. The study addresses six distinct postures: sleeping, sitting, standing, running, forward bending, and backward bending. The multi-layer perceptron and K-means algorithms were found to outperform the other algorithms. In another study, Van der Zwaard et al. [176] used a k-means clustering algorithm to classify athletes according to their individual anthropometric characteristics. All sprint cyclists were assigned to a mesomorphic group with endomorphy, mesomorphy, and ectomorphy values of 2.8, 5.0, and 2.4, respectively; the sample size was six. Pursuit and road cyclists were divided into a small meso-ectomorphic group with values of 1.6, 3.8, and 3.9; the sample size was nine, and a tall meso-ectomorphic group with values of 1.5, 3.6, and 4.0; the sample size was nine. Table 11 summarizes the above articles in the domain of HFE.

The K-Means clustering analysis facilitated the categorization of 29 human factors into four unique groups: subjective corrective accident human factors, perceived corrective accident human factors, associated corrective accident human factors, and critical corrective accident human factors. Following this classification, control recommendations were developed for each category in accordance with the results obtained [177].

VI. DISCUSSION

This review examines published literature on using ML algorithms in HFE from 2015 to 2024. Based on the articles included in the review, our findings cover various algorithms such as KNN, NB, decision trees, SVMs, RF, ANNs, DL, boosting algorithms, and k-means clustering algorithms. Articles were identified and categorized according to their relevance to the application areas involving these algorithms. This section provides a detailed discussion of the review results and their implications.

ML in HFE has been rapidly growing in popularity, attracting attention from researchers in recent years. Our study utilized a literature review to gain insight into the current landscape of ML studies within the realm of HFE, pinpointing existing research and highlighting areas that require additional investigation. WRMSDs and MWL assessment were the areas of focus of the selected studies. ML techniques, especially ANNs and DL, have demonstrated significant potential in advancing the field of WRMSDs and MWL assessment, offering more objective and real-time insights compared to traditional methods. Further research is required to enhance human-machine interaction, improve task efficiency, and promote user well-being in different fields. On the other hand, only two studies have utilized ML algorithms in occupational safety and health (OSH). ML can transform OSH practices by enabling proactive risk management, predictive analytics, and real-time monitoring. Future research applying ML to OSH holds immense potential to transform how workplaces manage risks and protect workers. By addressing these key areas, researchers can contribute to developing robust, adaptive, and ethically sound technologies that promote a safer and healthier work environment globally. In addition, ML offers advanced capabilities to analyze anthropometric data, extract patterns, and make predictions that enhance customization and usability in various

TABLE 10. Applications of boosting algorithms in the field of HFE (n=7).

Study	Aim	Application area, industry/sector
Chen et al. (2021) [171]	Examine the frequency of discomfort and pain due to musculoskeletal symptoms	WRMSDs and assessment of human postures, mining
Donisi et al. (2021) [170]	Categorize lifting activities into various lifting index categories based on the updated NIOSH lifting equation.	Manual material handling, wearable technology
Gutiérrez Teuler et al. (2022) [169]	Predict the workload of a controller by analyzing the controller's actions	Assessment of MWL, transportation
Harputlu Aksu and Çakıt (2023) [167]	Classifying MWL by using eye tracking data	Assessment of MWL, lab environment
Harputlu Aksu et al. (2024) [168]	Evaluate the assessment of MWL through the application of ML methods utilizing EEG and eye tracking data	Assessment of MWL, lab environment
Katić et al. (2020) [172]	Predict thermal comfort levels among individuals	Environmental ergonomics, lab environment
Nijeweme-d'Hollosy et al. (2018) [173]	Design a clinical decision support system to assist patients with LBP in self-referring to primary care	WRMSDs and assessment of human postures, healthcare

TABLE 11. Applications of k-means clustering algorithms in the field of HFE (n=3).

Study	Aim	Application area, industry/sector
Miao et al. (2023) [177]	Classify the 29 human factors into four distinct categories through the application of k-means clustering analysis.	HF, mining
Rababaah (2023) [175]	Classify human postures	WRMSDs and assessment of human postures, wearable technology
Van der Zwaard et al. (2019) [176]	Cluster athletes according to individual anthropometric characteristics	Anthropometric measurements, sports

domains. However, only three studies have used ML algorithms to analyze anthropometric data. By leveraging ML techniques effectively, researchers can unlock the full potential of anthropometric data to improve human-centered design and enhance quality of life.

ML is revolutionizing ergonomics by leveraging data-driven insights to enhance safety, comfort, and efficiency across various sectors including construction, education, healthcare, transportation, aviation, sports, mining, etc. From optimizing product design to improving workplace practices, ML offers valuable tools to create more ergonomic environments and solutions. However, integrating ML into HFE presents various ethical, data privacy, and practical deployment challenges that must be tackled to ensure responsible and effective implementation. Ethical concerns in ML applications within HFE primarily revolve around bias, fairness, accountability, and transparency. Since ML models are often trained on historical data, they may inherit societal or systemic biases. Additionally, many ML systems, particularly deep learning models, function as "black boxes," making it difficult to interpret their decision-making processes. In HFE, this lack of transparency can reduce user trust and complicate debugging or system improvement.

ML applications in HFE frequently rely on collecting and analyzing sensitive human data, including physiological signals, behavioral patterns, and workplace performance metrics. It is essential that users fully understand how their data is collected, stored, and utilized. Without proper consent mechanisms, ML in HFE could lead to unethical surveillance or manipulation. Beyond ethical and privacy concerns, real-world deployment of ML models in HFE presents technical and operational challenges. These models require large, high-quality datasets to function effectively, but obtaining such data can be difficult due to sensor limitations, human behavior variability, and environmental factors. Furthermore, ML models trained on controlled datasets often struggle to generalize to complex, real-world settings. For instance, an ML-powered ergonomic assessment tool trained in an office environment may not perform well in a factory or remote work setting. While ML has the potential to improve safety, efficiency, and usability in HFE, it also raises ethical concerns, privacy risks, and deployment difficulties. Addressing these challenges requires a multidisciplinary approach that includes regulatory oversight, strong security measures, and continuous human supervision to ensure that ML systems ethically and effectively serve human needs.

This review of the studies revealed several common issues that hinder the effective application of machine learning in HFE. One major issue is class imbalance, which occurs when there are significantly more examples of one outcome than another, such as when detecting rare but important events, such as injuries or high mental workload. Many studies have shown high overall accuracy, but they have not performed well in correctly identifying rare cases. Only a few studies employed helpful techniques, such as class weighting or balanced sampling, to address this problem. Ignoring class imbalance can lead to misleading results and make the model less reliable, especially in safety-related applications where catching rare cases is crucial.

Another common problem is the lack of external validation. Most studies tested their models using only the same data they were trained on, often from one source or lab setting. This makes it unclear whether the models will work well in real-world environments or with different groups of people. Very few studies tested their models on completely new data to check if the results would still hold. Additionally, the explainability of models is often lacking. Many researchers have used complex models, such as deep learning, but have not explained how these models make their decisions. This makes it hard for others to trust or improve the models. To make ML more effective in real-world HFE applications, researchers need to address these problems by employing more effective testing methods, handling data imbalance properly, and making models more comprehensible.

The presented review has several limitations related to the timeframe of the publications considered, thus potentially affecting study comprehensiveness, given that relevant articles published after the cutoff date (December 2024) were not included. A second limitation is that identifying relevant articles relied on a limited number of keywords, thus potentially leading to the omission of relevant publications using different terminology or keywords not considered in the search strategy. A third limitation is that article discovery was constrained by the databases chosen for the search. Although Web of Science, Science Direct, Google Scholar, IEEE Xplore, and Scopus are reputable sources, the lack of inclusion of other databases might have caused potentially relevant articles to be missed.

The exclusion of unrelated books, book chapters, conference proceedings, and review articles, although a practical choice for focusing the study, introduced a fourth limitation. Valuable information might exist in these formats, and their exclusion could have narrowed the scope of the review. Finally, the decision to include only full-text articles available in English limited the review's scope. Acknowledging these limitations is crucial for accurately interpreting the review findings. It highlights potential focus areas for future research efforts, such as expanding the search strategy to include additional databases, considering a broader range of publication types, and addressing language diversity in the inclusion criteria. In addition, there is a pressing need for future research to prioritize sectors that have not been thoroughly examined in previous research. This emphasis on unexplored sectors can motivate researchers to explore new avenues and make a significant impact in their field.

VII. CONCLUSION

This comprehensive review of 130 peer-reviewed journal articles from 2015 to 2024 highlights the growing integration of ML in the domain of HFE. The analysis revealed significant trends, notable algorithmic performance metrics, and key application domains within HFE. This literature review examined various machine learning (ML) algorithms, including KNN, NB, DT, SVM, RF, ANNs, DL, boosting algorithms, and k-means clustering, that have been applied in the HFE field over the last decade. A primary finding is the rising dominance of ANNs and DL techniques, which accounted for 36% (46 out of 130) of the reviewed studies. These models demonstrated superior performance in tasks such as mental workload assessment, posture recognition, and fatigue detection, achieving classification accuracies ranging from 88% to over 97%. SVMs, used in 22% of studies, also delivered high classification rates, particularly in applications involving posture and MWL classification. RF and KNN algorithms, while less frequently used, achieved competitive results in predictive ergonomics, with accuracies often exceeding 90%. From an application standpoint, the most common focus areas were WRMSDs (39% of studies), mental workload assessment (36%), and physical workload (10%). Notably, many ML models demonstrated potential for real-time implementation when combined with wearable sensors and computer vision systems, indicating readiness for workplace deployment.

The information provided herein indicates a substantial increase in published articles on ML algorithms applications, particularly after 2019. Nearly twice as many articles were published in 2020 than in 2019. Consequently, ML algorithms are increasingly being applied to solve various problems in the HFE domain. The presented review of ML applications in HFE should help to consolidate current knowledge, highlight best practices, address gaps, and guide future research. It should also support the development of new methods, frameworks, or best practices for applying ML in HFE. Future studies could build on this work by conducting more technically rigorous reviews or meta-analyses that delve deeply into the specific machine learning algorithms employed in HFE applications. Such investigations might systematically explore algorithmic configurations, hyperparameter optimization strategies, and the comparative impact of different model architectures (e.g., variations of SVM kernels, ANN layer designs, or ensemble methods). They could also critically examine common challenges such as overfitting in deep learning models, interpretability of complex algorithms in ergonomics contexts, and the influence of data imbalance or feature correlations on classification performance. By focusing on these technical dimensions, future research can provide valuable insights for ML specialists seeking to advance algorithm development and deployment within human factors and ergonomics.

Future research should prioritize the advancement of ML applications in HFE by investigating innovative algorithms, improving model interpretability, and overcoming existing challenges. The growing adoption of deep learning and artificial neural networks highlights the need for further studies on their practical applicability, scalability, and generalizability across various HFE domains. Moreover, incorporating explainable AI techniques could enhance trust and transparency in ML-driven decision-making. Future studies should also focus on developing standardized datasets and benchmarking frameworks to support model comparison and reproducibility. Additionally, fostering interdisciplinary collaboration between ML researchers and HFE practitioners may lead to more customized solutions that effectively address human-centered issues. Ethical considerations, such as data privacy and bias mitigation, should remain a key focus to ensure the responsible deployment of AI in HFE applications.

In conclusion, this review not only underscores the expanding role of ML in HFE but also establishes a performance-based benchmark for various algorithm types and applications. By addressing current limitations and following the outlined research directions, future studies can significantly advance both the scientific understanding and the practical impact of machine learning on the field of human factors and human-centered system design.

ACRONYMS

ANNs	Artificial Neural Networks.
DL	Deep Learning.
DT	Decision Trees.
ECG	Electrocardiogram.
EEG	Electroencephalography.
EMG	Electromyogram.
HF	Human Factors.
HFE	Human Factors and Ergonomics.
HRV	Heart Rate Variability.
KNN	K-Nearest Neighbors.
LBP	Low Back Pain.
ML	Machine Learning.
MWL	Mental Workload.
NB	Naive Bayes.
NN	Neural Networks.
PWL	Physical Workload.
RF	Random Forest.
SVM	Support Vector Machine.
OSH	Occupational Safety and Health.
XGBoost	Extreme Gradient Boosting.
WRMSDs	Work-Related Musculoskeletal Disorders.

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ERMAN ÇAKIT received the Ph.D. degree in industrial engineering from the University of Central Florida, Orlando, USA, in 2013. He is currently a Professor with the Department of Industrial Engineering, Gazi University, Türkiye. He has authored more than 50 papers published as peer-reviewed journal articles, book chapters, and conference proceedings. His research interests include applications of human factors/ergonomics and safety using machine learning and applied statistics. He serves as an Associate Editor for *Frontiers in Industrial Engineering*; and an Editorial Board Member for *Human-Intelligent Systems Integration* (Springer), *Augmented Human Research* (Springer), *International Journal of Machine Learning and Computing*, *Advances in Artificial Intelligence*, and *Machine Learning*.

WALDEMAR KARWOWSKI (Life Senior Member, IEEE) received the M.Sc. degree in production engineering from the University of Technology of Wroclaw, Poland, the Ph.D. degree in industrial engineering from Texas Tech University, USA, and the D.Sc. degree in management science from ORGMASZ, Poland. He is a Pegasus Professor and the Chair of the Department of Industrial Engineering and Management Systems; and the Executive Director with the Institute for Advanced Systems Engineering, University of Central Florida, Orlando, USA. His research interests include human systems integration, human factors, safety, neuroergonomics, soft computing, and nonlinear dynamics. He is a fellow of the Institute for Ergonomics and Human Factors Society (U.K.), the Human Factors and Ergonomics Society (HFES), the Institute of Industrial and Systems Engineers (IISE), the International Ergonomics Association (IEA), and American Association for Advancement of Science (AAAS). He received the William Floyd Award from the Chartered Institute of Ergonomics and Human Factors, U.K., in 2017; the David F. Baker Distinguished Research Award, in 2021; and the Albert Z. Holzman Distinguished Educator Award from the Institute of Industrial and Systems Engineers, USA, in 2025. He was elected to the Academy of Science, Engineering, and Medicine of Florida (ASEMFL), Orlando, FL, USA, in 2020. He was also awarded the Arnold M. Small and Betty Sanders President's Distinguished Service Award, HFES, in 2021. He was the past President of the Human Factors and Ergonomics Society, in 2007, and the International Ergonomics Association, from 2000 to 2003.